Project Proposal: CareBridge Khulna

Team: CSE4104-7B-T01 | Section: 7B | CSE 4104: Software Development III
GitHub: https://github.com/114nahid/cse4104-7b-t01-carebridge-khulna

1. Project Title

CareBridge Khulna: An AI-Powered Healthcare Assistance Platform for Elderly People

CareBridge Khulna is a web application that we are planning to build for elderly people living in Khulna city and for their family members who live far away from them. Through this system, a family member can find the right doctor for their parents, book a trusted university student to take them to the hospital and bring them back home, and follow every step of the visit from anywhere in the world. We will also use AI to make the whole process easier, especially for people who are not comfortable with technology.

2. Problem Statement

In Bangladesh, it has become very common for children to live away from their parents. Many young people from Khulna now live in Dhaka or other cities for their jobs, and a large number of them live abroad in countries like Saudi Arabia, Malaysia, the UK or the USA. Their parents usually stay back home in Khulna. As these parents get older, they need regular medical checkups for problems like diabetes, high blood pressure or heart disease. But going to the doctor is not easy for them. They have to arrange transport, wait for their serial, do medical tests, collect reports and then understand what the doctor has said. For an elderly person, doing all of this alone is tiring and sometimes not even safe.

The children who live far away feel this problem the most. They want their parents to stay healthy and to see a doctor whenever it is needed, but they have nobody to send with them. Sometimes a relative or a neighbour helps, but that is not possible every time. As a result, many elderly people delay their treatment, miss their follow-up visits, or go alone with a lot of difficulty. After the visit, the family usually gets only a short phone call, and they often do not clearly know what the doctor said or which medicines were given.

There are already some websites and apps in Bangladesh where people can search for doctors and book appointments. But these services stop at booking. Nobody goes with the patient to the hospital, nobody helps them inside the hospital, and the family does not get any update about what is happening. Families also often do not know which kind of specialist they should choose for a certain problem, and prescriptions are hard to read and understand. So the real problem, which is getting the elderly person to the doctor safely and keeping the family informed, is still not solved.

On the other side, there are thousands of university students in Khulna. Many of them come from middle-class families and look for part-time work to cover their own expenses. They are educated, they know the city well and they can easily handle hospital processes. If they are properly verified and managed, they can be the right people to accompany elderly patients. This gives students a respectful way to earn money and also gives families a person they can trust.

AI can make this system much more useful. A family member can simply describe the problem in Bangla, for example "my father feels pressure in his chest when he walks", and AI can suggest which type of specialist he should see. After the visit, AI can read the photo of the prescription and explain the medicines and their timing in simple Bangla. It can also turn the student's short notes into a proper visit report for the family. AI can even notice emergency words like "fainted" or "severe chest pain" and immediately show emergency contact numbers. These tasks are hard to do with normal programming, but they are a good fit for AI.

3. Objectives

The main objectives of our project are:

1. To help families, especially those living abroad or in other cities, arrange medical care for their elderly parents in Khulna from anywhere.
2. To make sure that elderly patients do not have to go to the hospital alone, by providing verified university students as hospital assistants.
3. To create a safe and organized part-time earning opportunity for university students.
4. To keep family members informed at every step of the hospital visit.
5. To use AI for suggesting the right specialist, explaining prescriptions in simple Bangla and preparing visit reports.
6. To build a simple system that works well on both mobile phones and computers, so that anyone can use it easily.

4. Proposed Features

Our system will have three types of users: family members, student assistants and the admin. Each user will log in with their own account and will see their own dashboard.

Family members. A family member will first create an account and add their parent's profile with basic information like age, address and existing health problems. They can also pin the home location on a map, so that the student can find the house easily. To find a doctor, they can search by specialty and area, or they can simply describe the problem in Bangla or English and let the AI suggest the right specialist. Searching by voice in Bangla will also be possible. After choosing a doctor and a date, they can book a hospital assistant. During the visit, they will see the status

change step by step on their dashboard, and after the visit they can read the prescription summary and the visit report. At the end, they can rate and review the student.

Student assistants. A student who wants to work with us will register with their university ID card and NID. They can start accepting visits only after the admin checks and approves their documents. When a visit is assigned, the student will go to the patient's home, take them to the hospital, stay with them during the visit and bring them back home. At each step, the student will update the status from their phone. After meeting the doctor, the student will upload a photo of the prescription and write a few short notes about what the doctor said.

Admin. The admin will verify new students, add and update doctor information, assign students to bookings and keep an eye on all bookings and reviews. Since we are not connecting directly with hospital systems, the admin or the student will confirm the doctor's serial by phone.

AI features. The AI part of the system will have four features. The *AI Specialist Finder* will understand the patient's problem from text or voice and suggest the right type of doctor. The *AI Prescription Summary* will read the prescription photo and explain the medicine names, their timing and the next visit date in simple Bangla. The *AI Visit Report* will turn the student's notes into a clear report for the family in Bangla or English. The *Emergency Alert* will detect serious symptoms and immediately show the 999 number and nearby emergency departments.

Besides these, the system will send notifications for bookings and status updates, and the whole website will be designed to work properly on mobile phones.

5. Technology Stack

We have chosen the following technologies for our project:

Part	Technology	Purpose
Frontend	React.js, Tailwind CSS	User interface and dashboards
Backend	Node.js, Express.js	REST APIs and business logic
Database	PostgreSQL	Storing users, doctors, bookings and reviews
Authentication	JWT, bcrypt	Secure login and password hashing
AI Integration	Gemini API	Specialist finder, prescription summary, visit report
Voice Search	Web Speech API	Bangla voice input
Image Storage	Cloudinary	Prescription photos and student documents

Part	Technology	Purpose
Map	Leaflet, OpenStreetMap, OpenRouteService	Home location, chamber location and distance
Deployment	Vercel, Render, Supabase	Hosting the frontend, backend and database
Other Tools	GitHub, Figma, Postman, Draw.io	Version control, UI design, API testing and diagrams

6. Expected Outcome

With CareBridge Khulna, a son or daughter living in Dubai or Dhaka will no longer feel helpless when their parents need a doctor. They will be able to arrange everything from their phone, knowing that a trusted person is with their parents and seeing what is happening at every step. After the visit, they will get a clear written report instead of an unclear phone call.

Elderly patients will not have to go to the hospital alone. Having someone with them will make the visit safer and less stressful, and it will also encourage them to go for regular checkups, so that health problems can be found early.

University students will get a chance to earn money while doing something meaningful for society. Because every student will be verified and rated by families, good students will build trust over time and get more work.

The AI features will save time in finding the right doctor, make medical information easy to understand for common people and give families a proper report without any extra work for the student. In an emergency, the system will help people take action faster. Overall, the system will turn a difficult and worrying process into a simple and organized one.

7. How the System Works

To explain how the system works, let us take a simple example. Rahim works in Dubai, and his mother lives in Sonadanga, Khulna. She needs to see a heart specialist, but Rahim cannot come home. He opens CareBridge Khulna, writes in Bangla that his mother feels chest pressure while walking, and the AI suggests that she should see a cardiologist. Rahim chooses a cardiologist from the list, picks a date and books a hospital assistant.

The admin assigns a verified student named Karim to this visit. On the visit day, Karim goes to the house and updates the status to "picked up". Rahim sees the update in Dubai. Then Karim updates "reached hospital" and "with doctor". After the checkup, Karim uploads a photo of the prescription and writes a few notes. The AI reads the prescription and explains in simple Bangla

which medicine to take and when. It also prepares a short visit report. When Karim brings Rahim's mother back home and updates "returned home", Rahim already knows everything that happened. At the end, he gives Karim a rating.

From the technical side, the React frontend will talk to the Express backend through REST APIs. The backend will check the user's login with JWT, save and read data from the PostgreSQL database, send requests to the Gemini API for the AI features, store images in Cloudinary and use OpenStreetMap for the map.

8. Project Scope

In this semester, we will build the complete system for the three types of users, the doctor directory, hospital assistance booking, student verification, step-by-step visit status, the four AI features, ratings and notifications. The website will be deployed online so that it can be used from anywhere.

To keep the project realistic for one semester, we will not build live GPS tracking, online payment, a direct connection with hospital systems, a separate login for doctors or a separate mobile app. Instead, the appointment will be confirmed by phone, payment will be made in cash, and the website itself will work on mobile phones. These features can be added in the future.

We also want to make it clear that the AI in our system will not diagnose any disease and will not suggest any medicine. It will only suggest the type of specialist and explain the prescription that a real doctor has written. This will be clearly mentioned inside the application.

9. Initial Project Plan

Week	Work	Deliverable
1	Team formation and idea selection	Team information and project idea
2	Project proposal	Proposal and GitHub repository
3	Requirement analysis	SRS document
4	System design	Architecture, ER diagram and API list
5	UI/UX design	Figma design
6	Backend development	Login, database and APIs
7	Frontend development	Pages connected with APIs
8	AI integration	AI features
9	Feature completion	Nearly complete system

Week	Work	Deliverable
10	Testing and bug fixing	Testing report
11	Deployment	Live website link
12	Documentation	Final report, user manual and API documentation
13	Presentation preparation	Presentation slides
14	Final presentation and viva	Live demo

10. Team Members and Responsibilities

Name	Student ID	Responsibility
M Sajjad Hossain (Team Leader)	11230121108	Database (PostgreSQL), AI integration (Gemini API), documentation and team coordination
Nahid Hasan Sabbir	11250122021	Backend (Node.js, Express.js), REST APIs and authentication
Md Sohanujjaman Sheikh	11230321265	Frontend (React.js, Tailwind CSS) and UI/UX design